

MECHANICAL & PLUMBING LEGEND

PLUMBING		PIPING		HVAC DUCTWORK	
	DOMESTIC COLD WATER		HWS — HEATING WATER SUPPLY		FIRE DAMPER
	DOMESTIC HOT WATER		HWR — HEATING WATER RETURN		SMOKE DAMPER
	DOMESTIC HOT WATER (TEMP. SHOWN)		CWS — CHILLED WATER SUPPLY		COMBINATION FIRE / SMOKE DAMPER
	DOMESTIC HOT WATER CIRCULATING		CWR — CHILLED WATER RETURN		TURNING VANES SHOWN IN 90° ELBOW.
	TEMPERED WATER		SMS — SNOW MELT SUPPLY		OPPOSED BLADE DAMPER (O.B.D.)
	NON-POTABLE WATER		SMR — SNOW MELT RETURN	DUCT SIZES ARE OUTSIDE SHEET METAL DIMENSIONS. 1st NO. IS SIZE OF SURFACE SHOWN. 2nd NO. IS DUCT DEPTH. LETTERS AFTER DIM. CORRESPOND TO FOLLOWING:	
	NATURAL GAS		RWS — RADIANT WATER SUPPLY		Ø - ROUND DUCTWORK
	PLUMBING VENT PIPE		RWR — RADIANT WATER RETURN		D - DOUBLE WALL CONSTRUCTION
	CHEMICAL VENT PIPE		D — DRAIN PIPE		UL - UNLINED DUCTWORK
	SANITARY WASTE PIPE		HPS — HIGH PRESSURE STEAM		V - OVAL DUCTWORK
	CHEMICAL WASTE PIPE		HPR — HIGH PRESSURE COND. RETURN		SUPPLY DIFFUSER. ALL DIFFUSERS FOUR-WAY THROW UNLESS NOTED OR THROWS SHOW IN PLAN. SEE SCHEDULE FOR TYPE.
	GREASE WASTE PIPE		LPS — LOW PRESSURE STEAM		RETURN AIR TAG. SEE SCHEDULE FOR TYPE.
	INDIRECT WASTE PIPE		LPR — LOW PRESSURE COND. RETURN		SECTION THRU OUTSIDE AIR INTAKE OR SUPPLY AIR DUCT
	SANITARY WASTE PIPE		CP — CONDENSATE PUMP DISCHARGE		SECTION THRU RETURN AIR OR EXHAUST DUCT
	SAND/OIL WASTE PIPE		FOS — FUEL OIL SUPPLY		OA, RA OR EXH DUCT DOWN
	INDIRECT DRAIN PIPE		FOR — FUEL OIL RETURN		SUPPLY AIR DUCT DOWN
	STORM DRAIN/ BUILDING STORM SEWER PIPE		FOV — FUEL OIL VENT		CHANGE OF ELEVATION UP (UP) DOWN (DN) IN DIRECTION OF AIR FLOW
	OVERFLOW STORM DRAIN PIPE		RS — REFRIGERANT SUCTION		FLEXIBLE CONNECTION
	VACUUM PIPE		RL — REFRIGERANT LIQUID		BACKDRAFT DAMPER
	COMPRESSED AIR PIPE		RD — REFRIGERANT DISCHARGE		RETURN AIR GRILLE W/SOUND BOOT SHADING INDICATES OPEN END
	COMPRESSED AIR LOW PRESSURE PIPE	VALVES & SPECIALTIES			FLEXIBLE DUCT
	COMPRESSED AIR HIGH PRESSURE PIPE		FLOW INDICATOR	CONTROLS (IN PLAN)	
	GEAR OIL PIPING		SHUT-OFF VALVE		THERMOSTAT/TEMP SENSOR
	WINDOW WASH FLUID		GLOBE VALVE		TEMP SENSOR W/STAINLESS STEEL COVER
	FUEL OIL SUPPLY		CHECK VALVE		TEMP SENSOR W/SETPOINT ADJUSTMENT
	FUEL OIL RETURN		FLOW CONTROL VALVE W/ CHECK		TEMP SENSOR W/TEMP INDICATION
	FUEL OIL VENT		CALIBRATED BALANCING SHUT-OFF VALVE.		TEMP SENSOR W/SET POINT ADJUSTMENT AND TEMP INDICATION
	PROPANE		PLUG VALVE.		THERMOSTAT/TEMP SENSOR WITH LOCKING GUARD
	W.C.O. WALL CLEANOUT		BALL VALVE		NIGHT THERMOSTAT
	F.C.O. FLOOR CLEANOUT		PRESSURE INDEPENDENT AUTOMATIC BALANCING VALVE		CARBON MONOXIDE SENSOR
	GCO GROUND CLEANOUT		N.O. VALVE W/ LOCKING COVER		CARBON DIOXIDE SENSOR
SNOWMELT			PLUG OR BALANCING SHUT-OFF VALVE IN RISER		HUMIDISTAT/SENSOR
	SNOWMELT SUPPLY		SHUT OFF VALVE IN RISER		STARTER
	SNOWMELT RETURN		DRAIN VALVE W/ HOSE END		PRESSURE SWITCH
MISCELLANEOUS			STRAINER W/ BLOW-OFF VALVE		DIFFERENTIAL AIR PRESSURE SENSOR
	POINT OF DUCT OR PIPE CONNECTION NEW TO EXISTING		TEMPERATURE CONTROL VALVE, 3-WAY		SWITCH
	POINT OF PIPE CONNECTION NEW TO EXISTING		TEMPERATURE CONTROL VALVE, 2-WAY		FLOW METER
	SECTION CUT: (A) SECTION I.D. (1) SHEET NO. WHERE SHOWN.		PRESSURE INDEPENDENT 2-WAY CONTROL VALVE		LOW LIMIT THERMOSTAT
	REFERENCE DESIGNATION FOR PLUMBING ISOMETRIC		PRESSURE REDUCING VALVE		TEMPERATURE CONTROL (T.C.) DAMPER
	REFERENCE DESIGNATION FOR COMPRESSED AIR DETAIL. WHERE MANIFOLD DETAIL IS REFERED THE # OF OUTLETS WILL BE NOTED, COORDINATE EXACT LOCATION WITH EQUIPMENT. SEE COMPRESSED AIR DETAILS ON SHEET P503.		SAFETY OR RELIEF VALVE		SMOKE DETECTOR (DUCT)
			MANUAL AIR VENT		PRESSURE DIFFERENTIAL SENSOR
			AUTOMATIC AIR VENT		SECTION CUT: (A) SECTION I.D. (1) SHEET NO. WHERE SHOWN.
			PRESSURE - TEMP. TAP		UNDERCUT DOOR, 1".
			PRESSURE GAUGE W/ PIG TAIL & COCK		REFERENCE DESIGNATION FOR PLUMBING ISOMETRIC
			THERMOMETER		
			VACUUM BREAKER		
			PIPE EXPANSION JOINT		
			FLEXIBLE PIPE CONNECTOR (STAINLESS STEEL BRAIDED)		
			FLEXIBLE EQUIPMENT CONNECTOR (RUBBER DOUBLE BELLOWS)		
			PIPE UNION		
			PIPE CAP		
			PIPE ANCHOR		
			PIPE GUIDE		
			PREMANUFACTURED PIPE EXPANSION LOOP. REQUIRED EXPANSION COMPENSATION LISTED. PROVIDE GUIDES & ANCHORS AS RECOMMENDED BY MANUFACTURER.		
		MISCELLANEOUS			
			POINT OF DUCT OR PIPE CONNECTION NEW TO EXISTING		
			POINT OF PIPE CONNECTION NEW TO EXISTING		

ABBREVIATIONS

AFF	ABOVE FINISHED FLOOR
AFG	ABOVE FINISHED GRADE
ALT	ALTITUDE
BHP	BRAKE HORSEPOWER
BTU	BRITISH THERMAL UNIT
C	COMMON PORT
CFM	CUBIC FEET PER MINUTE
SCFM	CFM, STANDARD CONDITIONS
CU FT	CUBIC FEET
CU IN	CUBIC INCH
Cv	COEFFICIENT, VALVE FLOW
dB	DECIBEL
DB	DRY BULB
DCW	DOMESTIC COLD WATER
DEG OR °	DEGREE
DHW	DOMESTIC HOT WATER
DIA	DIAMETER
DN	DOWN
(E)	EXISTING
EAT	ENTERING AIR TEMP.
EFF	EFFICIENCY
ELEV	ELEVATION
ESP	EXTERNAL STATIC PRESSURE
EW	ENTERING WATER TEMP.
EXH	EXHAUST
F	FAHRENHEIT
LF	LINEAR FEET
LRA	LOCKED ROTOR AMPS
LWT	LEAVING WATER TEMPERATURE
MAX	MAXIMUM
MBH	BTU PER HOUR (THOUSAND)
MCA	MINIMUM CIRCUIT AMPS
MIN	MINIMUM
N/A	NOT APPLICABLE
N.C.	NORMALLY CLOSED
N.O.	NORMALLY OPEN
NC	NOISE CRITERIA
NIC	NOT IN CONTRACT
NUMBER	NUMBER
NO OR #	NOT TO SCALE
NTS	NOT TO SCALE
OA	OUTSIDE AIR
OD	OUTSIDE DIAMETER
%	PERCENT
PH OR Ø	PHASE (ELECTRICAL)
PPM	PARTS PER MILLION
PRESS	PRESSURE
PSF	POUNDS PER SQUARE FOOT
PSI	POUNDS PER SQUARE INCH
PSIA	PSI ABSOLUTE
PSIG	PSI GAUGE
RA	RETURN AIR
RECIRC	RECIRCULATE
FLA	FULL LOAD AMPS
FPM	FEET PER MINUTE
FPS	FEET PER SECOND
FT	FOOT, FEET, FEET WATER COLUMN
FU	FIXTURE UNITS
GA	GAUGE
GAL	GALLONS
GPH	GALLONS PER HOUR
GPM	GALLONS PER MINUTE
HD	HEAD
HG	MERCURY
HGT	HEIGHT
HORZ	HORIZONTAL
HP	HORSEPOWER
HR	HOUR(S)
HWC	HOT WATER CIRCULATING (DOMESTIC)
HZ	HERTZ
ID	INSIDE DIAMETER
IE	INVERT ELEVATION
IJS	IN JOIST SPACE
IN	INCHES
IN W.C.	INCHES IN INCHES WATER COLUMN
KW	KILOWATT
KWH	KILOWATT HOUR
LAT	LEAVING AIR TEMPERATURE
LBS OR #	POUNDS
RH	RELATIVE HUMIDITY
RLA	RUNNING LOAD AMPS
RPM	REVOLUTIONS PER MINUTE
SA	SUPPLY AIR
SENS	SENSIBLE
SL	SEA LEVEL
SP	STATIC PRESSURE
SPEC	SPECIFICATION
SQ	SQUARE
STD	STANDARD
TD	TEMPERATURE DIFFERENCE
TEMP	TEMPERATURE
THRU	THROUGH
TJO	THROUGH JOIST OPENINGS
TONS	TONS OF REFRIGERATION
TSP	TOTAL STATIC PRESSURE
TSTAT	THERMOSTAT
TYP	TYPICAL
V	VOLT
VAV	VARIABLE AIR VOLUME
VEL	VELOCITY
VERT	VERTICAL
VOL	VOLUME
W	WATT
WB	WET-BULB
WPD	WATER PRESSURE DROP
WT	WEIGHT

PROJECT INFORMATION

PROJECT LOCATION:	BOULDER, COLORADO
PROJECT ALTITUDE:	5,400 FEET ABOVE SEA LEVEL
SUMMER DESIGN TEMPERATURE:	94°F DRY BULB, 61°F WET BULB (ASHRAE 0.4% DESIGN)
WINTER DESIGN TEMPERATURE:	-4°F DRY BULB (ASHRAE 99.6%)
SUMMER ROOM DESIGN TEMP:	75°F DRY BULB - NO HUMIDITY REQUIREMENT
WINTER ROOM DESIGN TEMP:	72°F DRY BULB - NO HUMIDITY REQUIREMENT

DRAWING INDEX

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GENERAL NOTES (FOR ALL MECHANICAL DRAWINGS)

- INFORMATION ON THE DRAWINGS HAS BEEN ASCERTAINED FROM EXISTING DRAWINGS AND FIELD OBSERVATIONS. THIS INFORMATION IS AS ACCURATE AS CONDITIONS WOULD ALLOW. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO VISIT THE SITE, PRIOR TO BID, AND BECOME FAMILIAR WITH THE EXTENT OF THE WORK REQUIRED. NO EXTRAS WILL BE ALLOWED FOR ALTERATIONS OF A FORESEEABLE NATURE REQUIRED TO ACHIEVE THE END RESULT AS INDICATED BY CONTRACT DOCUMENTS.
- CONTRACTOR IS TO PROVIDE COMPLETE CONNECTIONS TO ALL EQUIPMENT.
- CONTRACTOR SHALL MAKE ARRANGEMENTS FOR PARKING, COORDINATE ANY EQUIPMENT DOWNTIME, AND COORDINATE GENERAL WORK SCHEDULE WITH OWNER.
- CONTRACTOR SHALL VISIT THE SITE BEFORE SUBMITTING BID, AND VERIFY DIMENSIONS AND EXISTING CONDITIONS RELATED TO THE WORK.
- ALL CUTTING, PATCHING AND CORE DRILLING FOR THE INSTALLATION OF NEW EQUIPMENT, DUCTS, HANGERS, ETC. SHALL BE HELD TO A MINIMUM AND BE ACCOMPLISHED IN A CAREFUL MANNER.
- CONTRACTOR TO COORDINATE THE LOCATION OF ALL DUCTWORK, CEILING DEVICES, GRILLES, REGISTERS AND DIFFUSERS WITH REFLECTED CEILING PLAN AND STRUCTURE PRIOR TO BEGINNING WORK.
- DUCT DIMENSIONS GIVEN ARE OUTSIDE SHEET METAL DIMENSIONS.
- MECHANICAL DRAWINGS ARE DIAGRAMMATIC AND DO NOT NECESSARILY INDICATE EVERY REQUIRED OFFSET, FITTING, ETC. DRAWINGS ARE NOT TO BE SCALED FOR DIMENSIONS. TAKE ALL DIMENSIONS FROM ARCHITECTURAL DRAWINGS, CERTIFIED EQUIPMENT DRAWINGS AND FROM THE STRUCTURE ITSELF BEFORE FABRICATING ANY WORK. VERIFY ALL SPACE REQUIREMENTS COORDINATING WITH OTHER TRADES, AND INSTALL THE SYSTEMS IN THE SPACE PROVIDED WITHOUT EXTRA CHARGES TO THE OWNER.
- THE OWNER AND ENGINEER ARE NOT RESPONSIBLE FOR THE CONTRACTOR'S SAFETY PRECAUTIONS OR TO MEANS, METHODS, TECHNIQUES, CONSTRUCTION SEQUENCES, OR PROCEDURES REQUIRED TO PERFORM HIS WORK.
- ALL WORK SHALL BE INSTALLED IN ACCORDANCE WITH ALL APPLICABLE STATE CODES, LOCAL CODES AND OWNER'S STANDARDS INDICATED BY THE CONSTRUCTION DOCUMENTS.
- ALL EXTERIOR WALL AND ROOF PENETRATIONS SHALL BE SEALED WATERPROOF.
- ALL MECHANICAL WORK UNDER THIS CONTRACT IS TO FIVE (5) FEET OUTSIDE THE BUILDING.
- XX-## REFERS TO EQUIPMENT DESIGNATION. SEE SPECIFICATIONS AND EQUIPMENT SCHEDULES FOR FURTHER INFORMATION IN REFERENCE TO THAT DESIGNATED PIECE OF EQUIPMENT.
- DRAWINGS INDICATE UNITS IN IP UNITS. WHERE SUBMITTED MATERIAL USES METRIC UNITS, SUCH AS PRE-INSULATED PEX, THE SUBMITTED MATERIAL INSIDE FREE AREA SHALL BE EQUIVALENT OR LARGER THAN THE IDENTIFIED IP PIPE SIZE FREE AREA.
- CONTRACTOR SHALL FIELD COORDINATE EXACT LOCATION OF RADIANT / SNOWMELT PEX TUBING WITH STRUCTURAL / ARCHITECTURAL DRAWINGS, INDUSTRIAL EQUIPMENT DRAWINGS, AND GENERAL CONTRACTOR PRIOR TO CONCRETE POUR.

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY
BOULDER DESIGN AND CONSTRUCTION DIVISION
325 BROADWAY BOULDER, COLORADO 80305-3328

BUILDING 42 TUNNEL VENTILATION
MECHANICAL LEGENDS AND
GENERAL NOTES

DESIGN REESE	DRAFTING COOK	APPROVAL	DATE 4/1/2026	SHEET NUMBER M-0 of
PROJECT NAME BUILDING 42 TUNNEL VENTILATION				

REVISION	DESCRIPTION	BY	SUPV.	DATE	